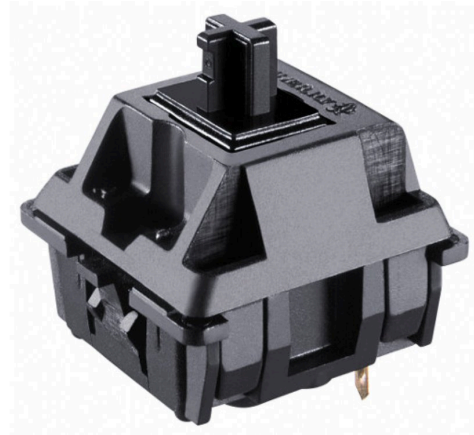


Bill of Materials

Images	Description	Quantity/Weight
	PLA 3D printing filament	1kg
	Spool of 22 Gauge Wire	1 foot

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	Male-to-Female Jumper Wires	12
	Cherry MX Black Switch Keys	12
	Arduino Pro Micro	1



USB-C to Micro-USB Connector Wire

1

Assembly Process

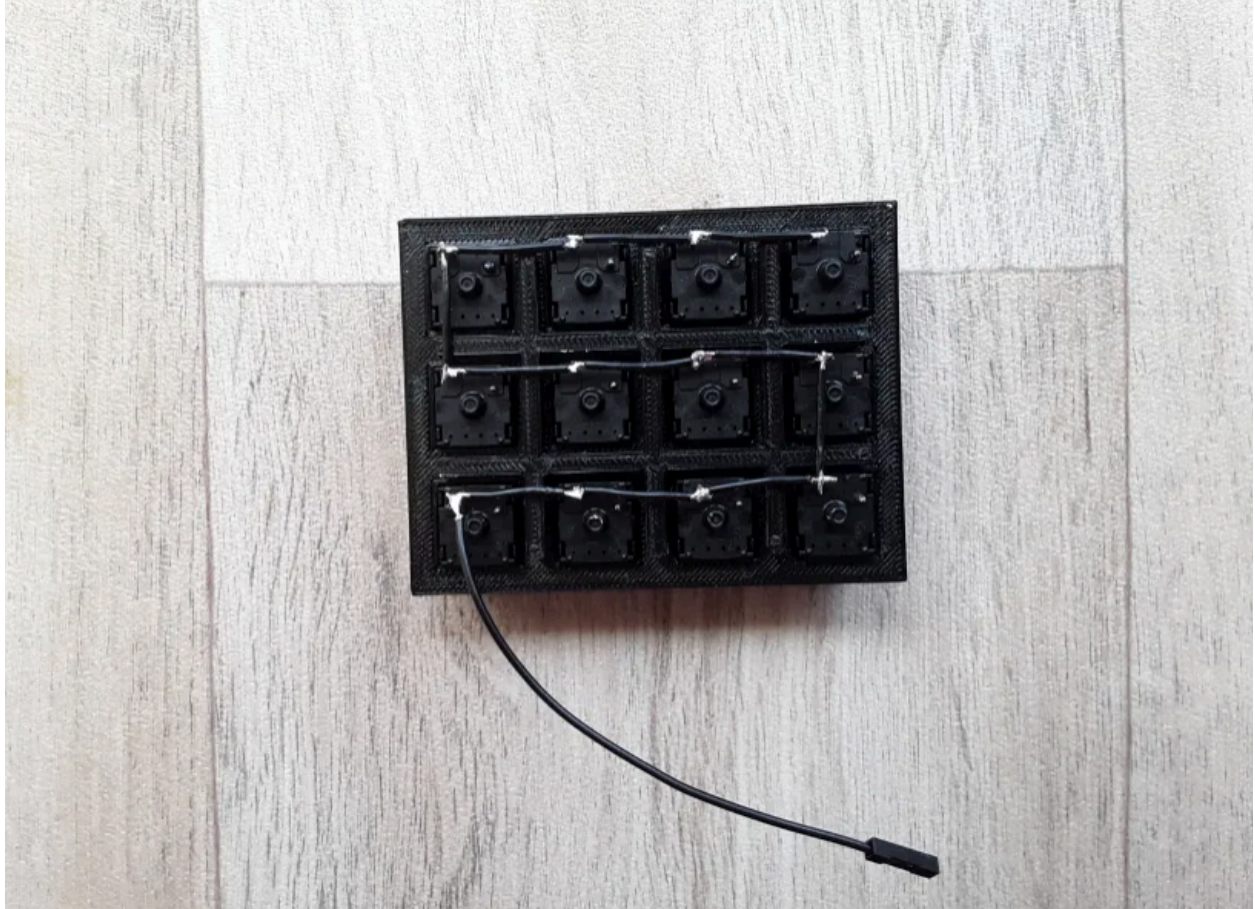
Building the TouchType

Step 1: Download and print the TouchType Main Body and Keycaps.

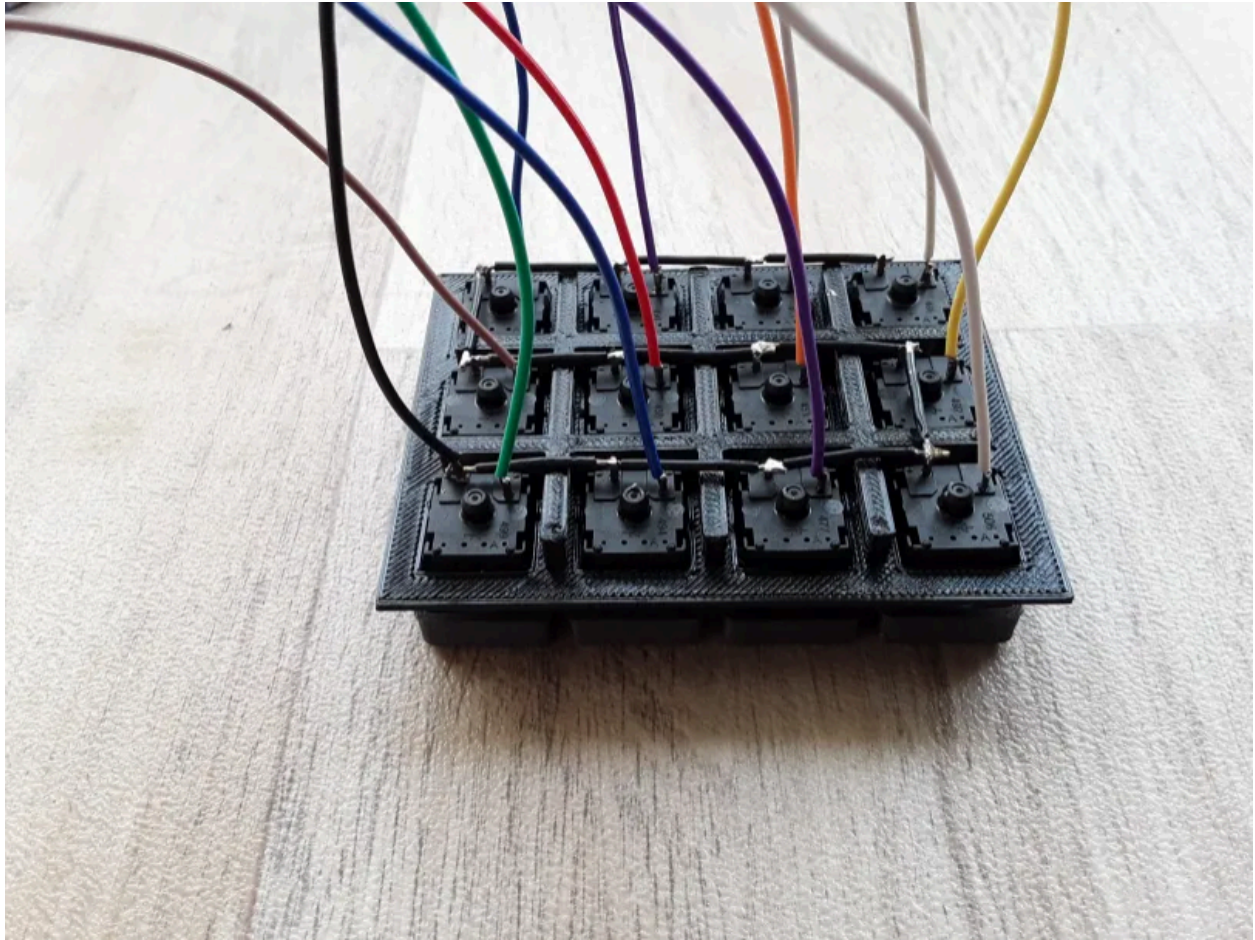
Step 2: Insert Arduino Pro Micro into the Arduino housing in the TouchType Main Body.

Step 3: Insert the 12 Cherry MX Black Switches into the switch grid.

Step 4: Solder 22 gauge wiring to each ground output of each Cherry switch on the opposite side of the switch grid that has the outward facing walls.

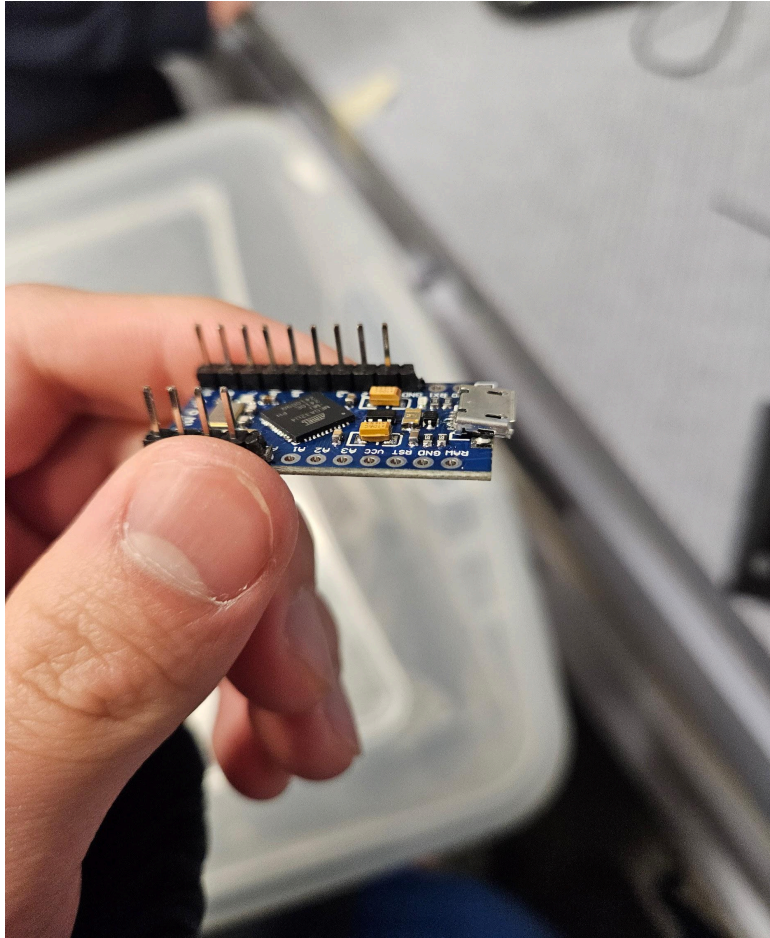


Step 5: Solder each switch's input to a jumper wire.

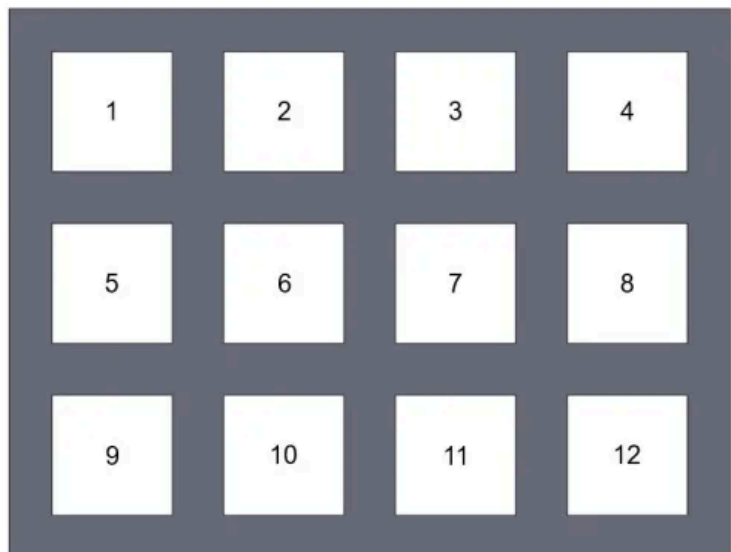
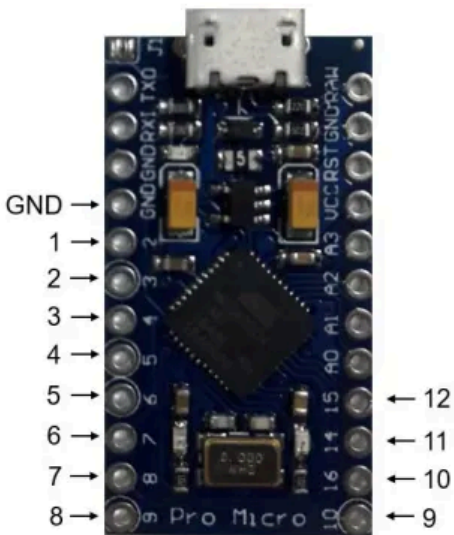


Step 6: Solder header pins onto the Arduino Pro Micro in this orientation.

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Step 7: Connect each switch's jumper wire to the Arduino Pro Micro as seen in this photo:



Step 8: Connect Arduino Pro Micro to a computer via the Micro-USB C connector cable.

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Step 9: Upload the code for the Arduino Pro Micro. Check to make sure all the inputs are correct and each switch is connected.

Step 10: Secure switch grid onto the TouchType main body.

Step 11: Insert Braille Indents into holes of keycaps, use a clamp to squeeze the two together, the indents should stick out of the keycaps slightly in order to keep the tactile feeling of the keycaps.

Step 12: Secure the keycaps onto the Cherry Keycaps.



Step 13: The TouchType is now complete!